DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs to improve slot-filling and tool call rejection without human labeling.

arxiv.org/abs/2504.02882

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must control conversation flow by deciding whether to ask follow-up questions, make a tool call, or reject inappropriate requests. SFT with expert trajectories alone has critical limitations.

Failure to Ask Follow-Up Questions

When queries lack required arguments, SFT-trained models skip slot-filling and hallucinate parameter values.

Hallucinated Tool Invocations

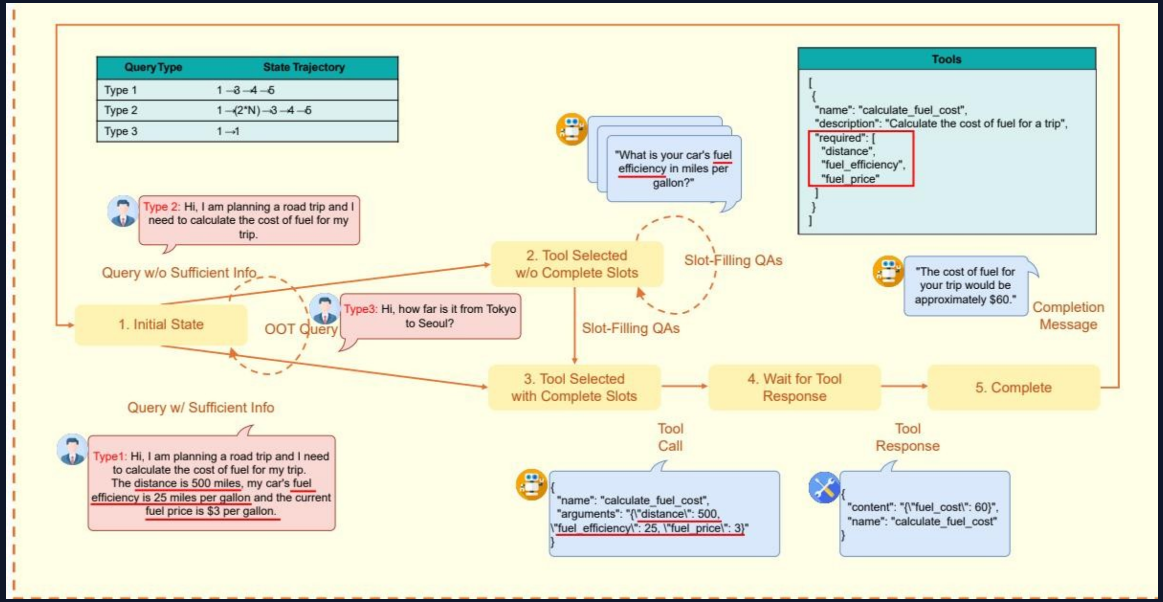
Models trained on expert trajectories overgeneralize, invoking tools even when information is incomplete.

Failure to Reject Out-of-Scope Requests

Without preference signal, models cannot learn to reject queries for which no suitable tool exists.

Solution: Direct Preference Optimization with automatically constructed paired trajectory datasets of correct vs. incorrect dialogue flows.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



S1

Initial

No history. Out-of-scope requests return rejection and remain here.

S2

Tool Selected w/o Complete Slots

Tool identified but required parameters are missing. Slot-filling QA occurs.

S3

Tool Selected w/ Complete Slots

Both tool selection and all argument values are determined.

S4

Wait for Tool Response

Tool call executed, awaiting results from the tool.

S5

Complete

Tool results received, completion message generated for the user.

Three Query Types Define Distinct State Trajectories

User queries are classified into three types based on their state transition trajectories:

Query Type	State Trajectory	Description	Example Scenario
Type 1	1 → 3 → 4 → 5	All required arguments present. No slot-filling needed.	"Book a flight from NYC to LAX on Dec 15" → All slots filled, invoke tool directly
Type 2	1 → (2×N) → 3 → 4 → 5	Missing arguments. Slot-filling QA repeats N times.	"Book a flight" → Ask: destination? date? → Fill slots → Invoke
Type 3	1 → 1	Out-of-scope request. Tool call must be rejected.	"Write me a poem about the sunset" → No suitable tool → Reject & stay in S1

Key Insight: These three trajectory patterns form the basis for automatic preference pair construction — each query type defines what the chosen (correct) and rejected (incorrect) dialogue flows look like.

Automatic Paired Trajectory Construction Without Human Labels

Chosen vs. rejected trajectories are paired automatically for each query type, teaching specific dialogue control lessons.

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	3,092	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,652	Easy/Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	3,092	Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

All pairs constructed automatically from state trajectories — zero human annotation required.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

FunctionChat-Bench evaluation results

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	**0.833**	0.609	0.575	0.614
Prop.-8B	SFT-only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	**0.957**	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

Green = best in model group | Highlighted rows = SFT + DiaTool-DPO (combined approach)

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Slot Accuracy

0.639 → **0.917**

+43.5%

Relevance

0.826 → **0.913**

+10.5%

Macro Average

0.816 → **0.904**

+10.8%

Approaches near GPT-4o performance: 94.8% in information gathering, 91% in tool call rejection.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1. **MDP formulation enables systematic preference pair construction**

   Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

2. **Automatic dataset construction eliminates human labeling**

   Paired trajectories are derived directly from state-transition patterns, making the approach scalable and reproducible.

3. **DPO complements SFT — it does not replace it**

   DiaTool-DPO-only performs poorly across all models. Combined SFT + DiaTool-DPO consistently achieves the best Macro Average.

4. **Slot-filling and tool rejection see the largest gains**

   LLaMA-3-8B: Slot accuracy +43.5% (0.639 → 0.917), Relevance +10.5% (0.826 → 0.913) over SFT-only.

5. **Language-agnostic and production-ready**

   Primary experiments in Korean generalize to English — opens broad applicability for production TA-LLMs at near GPT-4o performance levels.

Source: arxiv.org/abs/2504.02882